

AWT LAB EXERCISE:

PART A

Exercise 1: Create a page with Tailwind CSS product cards styled using `max-w-sm`, `shadow-md`, `rounded-lg`, and hover effects like `hover:shadow-xl`. Style the image with `object-cover`, `rounded-t-lg`, `h-48` and text with `font-bold`, `text-gray-700`, and `text-sm`.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Tailwind CSS Cards</title>
  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-gray-100 p-10">
  <!-- Responsive grid container -->
  <div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-6">
    <!-- Card 1 -->
    <div class="shadow-md rounded-lg bg-white">
      
      <div class="p-4">
        <h2 class="font-bold text-gray-700">Product Name 1</h2>
        <p class="text-gray-600 text-sm mt-2">
          This is a description of the product.
        </p>
        <div class="mt-4">
          <span class="font-bold text-gray-800">Rs.350</span>
          <button class="ml-4 bg-blue-500 text-white text-sm py-1 px-3 rounded hover:bg-red-600">
            Buy Now
          </button>
        </div>
      </div>
    </div>

    <!-- Card 2 -->
    <div class="shadow-md rounded-lg bg-white">
      
      <div class="p-4">
        <h2 class="font-bold text-gray-700">Product Name 2</h2>
        <p class="text-gray-600 text-sm mt-2">
```

This is a description of the product.

</p>

<div class="mt-4">

Rs.200

<button class="ml-4 bg-blue-500 text-white text-sm py-1 px-3 rounded

hover:bg-blue-600">

Buy Now

</button>

</div>

</div>

</div>

<!-- Card 3 -->

<div class="shadow-md rounded-lg bg-white">

<div class="p-4">

<h2 class="font-bold text-gray-700">Product Name 3</h2>

<p class="text-gray-600 text-sm mt-2">

This is a description of the product.

</p>

<div class="mt-4">

Rs.400

<button class="ml-4 bg-blue-500 text-white text-sm py-1 px-3 rounded

hover:bg-blue-600">

Buy Now

</button>

</div>

</div>

</div>

<!-- Card 4 -->

<div class="shadow-md rounded-lg bg-white">

<div class="p-4">

<h2 class="font-bold text-gray-700">Product Name 4</h2>

<p class="text-gray-600 text-sm mt-2">

This is a description of the product.

</p>

<div class="mt-4">

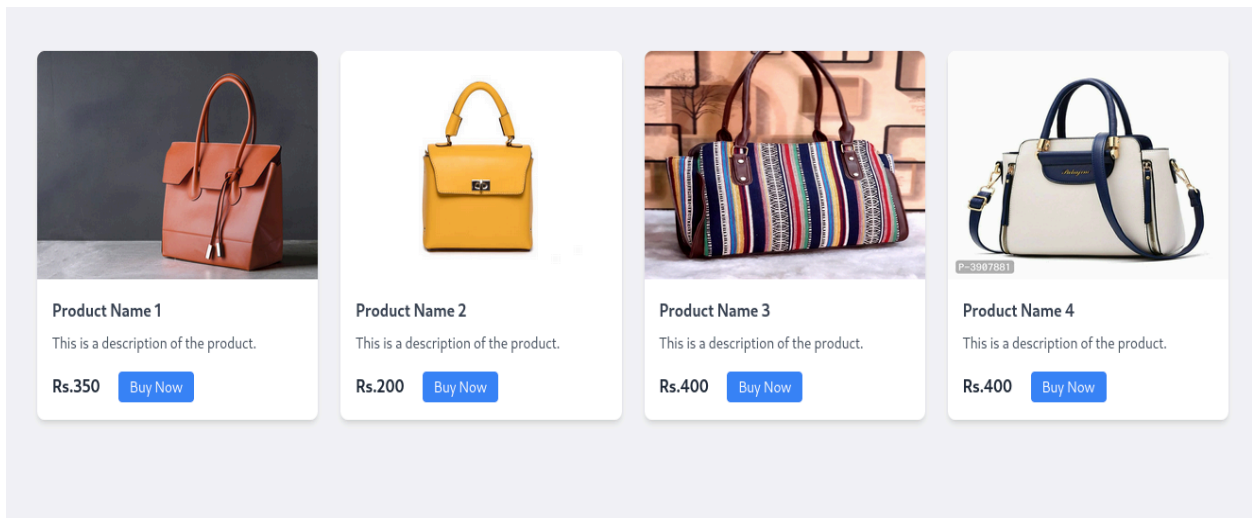
Rs.400

```

        <button class="ml-4 bg-blue-500 text-white text-sm py-1 px-3 rounded
hover:bg-blue-600">
        Buy Now
    </button>
</div>
</div>
</div>
</div>
</div>
</body>
</html>

```

OUTPUT:



EXERCISE 2: Create a developer portfolio website using Bootstrap with a responsive navbar, a hero section featuring display-4 text and a btn-primary button, and a projects section using a grid (row and col-md-4) to display three card components. Include a contact form with form-group fields for name, email, and message, and finish with a footer featuring social media icons.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Personal Portfolio</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    /* Custom styles for a modern color scheme */
    body {
      background-color: #f9f9f9; /* Light Gray */
      color: #333333;
      font-family: 'Arial', sans-serif;
    }

    .navbar {
      background-color: #000000; /* Teal */
    }

    .navbar-brand, .nav-link {
      color: #fff !important;
      font-weight: bold;
    }

    .navbar-toggler-icon {
      background-color: #fff;
    }

    .hero-section {
      background-color: #008080; /* Coral */
      color: #fff;
      padding: 100px 0;
    }

    .hero-section h1 {
```

```
font-size: 3rem;  
font-weight: bold;  
}
```

```
.hero-section p {  
font-size: 1.25rem;  
}
```

```
.section-title {  
color: #008080; /* Teal */  
font-weight: bold;  
margin-bottom: 30px;  
}
```

```
.card {  
border: none;  
border-radius: 15px;  
box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);  
}
```

```
.card img {  
border-radius: 15px 15px 0 0;  
}
```

```
.card-body {  
padding: 20px;  
}
```

```
.card-title {  
font-weight: bold;  
}
```

```
.footer {  
background-color: #333333; /* Dark Gray */  
color: #fff;  
padding: 20px 0;  
text-align: center;  
}
```

```
.btn-primary {  
background-color: #000000; /* Teal */  
border: none;  
}
```

```
.btn-primary:hover {  
background-color: #006f66; /* Darker Teal */  
}
```

```
.social-icons i {  
font-size: 1.5rem;  
margin-right: 15px;  
color: #008080;  
transition: all 0.3s ease;  
}
```

```
.social-icons i:hover {  
color: #006f66;  
}
```

```
.contact-form input,  
.contact-form textarea {  
border-radius: 10px;  
}
```

```
.contact-form button {  
border-radius: 10px;  
}
```

```
</style>  
</head>  
<body>
```

```
<!-- Navbar -->  
<nav class="navbar navbar-expand-lg">  
  <div class="container">  
    <a class="navbar-brand" href="#">SHREYA K</a>  
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse"  
data-bs-target="#navbarNav">  
      <span class="navbar-toggler-icon"></span>  
    </button>  
    <div class="collapse navbar-collapse" id="navbarNav">  
      <ul class="navbar-nav ms-auto">  
        <li class="nav-item"><a class="nav-link" href="#about">About</a></li>  
        <li class="nav-item"><a class="nav-link" href="#projects">Projects</a></li>  
        <li class="nav-item"><a class="nav-link" href="#contact">Contact</a></li>  
      </ul>  
    </div>  
  </div>  
</nav>
```

```

<!-- Hero Section -->
<header class="hero-section text-center">
  <div class="container">
    <h1>Hi, I'm Shreya K</h1>
    <p class="lead">A passionate developer specializing in modern web technologies.</p>
    <a href="#contact" class="btn btn-primary">Get in Touch</a>
  </div>
</header>

<!-- About Section -->
<section id="about" class="py-5">
  <div class="container">
    <h2 class="section-title text-center">About Me</h2>
    <p class="text-center">I'm a dedicated developer with experience in building responsive and
scalable web applications. I specialize in modern web technologies such as React, Python, and Machine
Learning. I love tackling new challenges and continuously learning and growing in my field.</p>
  </div>
</section>

<!-- Projects Section -->
<section id="projects" class="py-5 bg-light">
  <div class="container">
    <h2 class="section-title text-center">Projects</h2>
    <div class="row">
      <div class="col-md-4">
        <div class="card">
          
          <div class="card-body">
            <h5 class="card-title">Project 1</h5>
            <p class="card-text">This project is a web application that allows users to track their daily tasks and
goals.</p>
            <a href="#" class="btn btn-primary">View Project</a>
          </div>
        </div>
      </div>
      <div class="col-md-4">
        <div class="card">
          
          <div class="card-body">
            <h5 class="card-title">Project 2</h5>
            <p class="card-text">A mobile-responsive portfolio website showcasing my skills and projects.</p>
            <a href="#" class="btn btn-primary">View Project</a>
          </div>
        </div>
      </div>
    </div>
  </div>

```

```
    </div>
  </div>
  <div class="col-md-4">
    <div class="card">
      
      <div class="card-body">
        <h5 class="card-title">Project 3</h5>
        <p class="card-text">A machine learning model that predicts future trends based on historical
data.</p>
        <a href="#" class="btn btn-primary">View Project</a>
      </div>
    </div>
  </div>
</section>
```

<!-- Contact Section -->

```
<section id="contact" class="py-5">
  <div class="container">
    <h2 class="section-title text-center">Contact Me</h2>
    <form class="contact-form">
      <div class="row">
        <div class="col-md-6 mb-3">
          <input type="text" class="form-control" placeholder="Your Name" required>
        </div>
        <div class="col-md-6 mb-3">
          <input type="email" class="form-control" placeholder="Your Email" required>
        </div>
        <div class="col-md-12 mb-3">
          <textarea class="form-control" placeholder="Your Message" rows="5" required></textarea>
        </div>
        <div class="col-md-12 text-center">
          <button type="submit" class="btn btn-primary">Send Message</button>
        </div>
      </div>
    </form>
  </div>
</section>
```

<!-- Footer -->

```
<footer class="footer">
  <p>&copy; 2025.All Rights Reserved.</p>
```


</footer>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/js/bootstrap.bundle.min.js"></script>

</body>

</html>

EXERCISE:3 Create a React application that includes a form with a textbox component, a dropdown component, and a submit button component. Each form element should be implemented as a separate, reusable component.

Dropdown.js

```
// src/components/Dropdown.js
import React from 'react';

function Dropdown() {
  return (
    <div className="form-group">
      <label>Select Option</label>
      <select>
        <option>MCA</option>
        <option>MBA</option>
        <option>Msc</option>
        <option>MCom</option>
        <option>Mtech</option>
      </select>
    </div>
  );
};

export default Dropdown;
```

Textbox.js

```
// src/components/Textbox.js
import React from 'react';

function Textbox() {
  return (
    <div>
      <label>Enter the Text</label>
      <input
        type="text"
      >
    </div>
  );
};

export default Textbox;
```

```
    />
  </div>
);
};

export default Textbox;
```

Submit.js

```
import React from 'react';

function Submit() {
  return (
    <button>
      Submit
    </button>
  );
};

export default Submit;
```

App.js

```
import React from 'react';
import Textbox from './Textbox';
import Dropdown from './Dropdown';
import Submit from './Submit.js';
function App() {
  return (
    <> <h1>Simple Form</h1>
      <Textbox/><br></br>
      <Dropdown/><br></br>
      <Submit/></>
    </>
  );
};

export default App;
```

```
);  
};  
  
export default App;
```

OUTPUT:

Simple Form

Enter the Text

Select Option

Simple Form

Enter the Text

Select Option

- MCA
- MBA
- Msc
- MCom
- Mtech

EXERCISE:4 Build a React application that includes a form to accept a first name and a last name using Material UI components. Instead of a standard “Submit” button, implement a “Greet Me” button that, when clicked, will display an alert with the message “Hello [first name] [last name]!”.

App.js

```
import React, { useState } from "react";
import TextField from "@mui/material/TextField";
import Button from "@mui/material/Button";

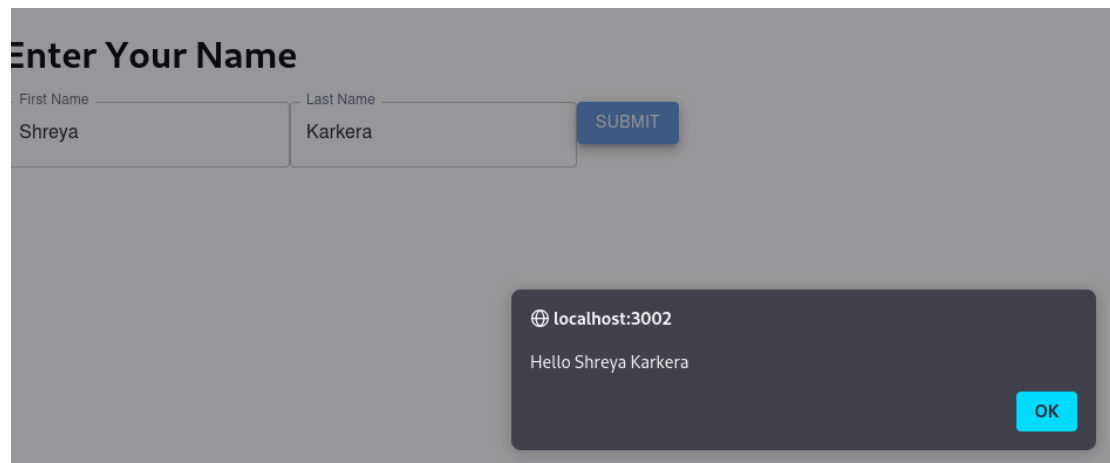
function App() {
  const [firstName, setFirstName] = useState("");
  const [lastName, setLastName] = useState("");

  const handleClick = () => {
    if (firstName === "" || lastName === "") {
      alert("Please fill out both fields.");
    } else {
      alert(`Hello {firstName} ${lastName}`);
    }
  };

  return (
    <div>
      <h1>Enter Your Name</h1>
      <TextField
        label="First Name"
        variant="outlined"
        value={firstName}
        onChange={(e) => setFirstName(e.target.value)}
      />
      <TextField
        label="Last Name"
        variant="outlined"
        value={lastName}
        onChange={(e) => setLastName(e.target.value)}
      />
      <Button variant="contained" color="primary" onClick={handleClick}>
```

```
        Submit  
      </Button>  
    </div>  
  );  
}  
  
export default App;
```

OUTPUT:



The screenshot displays a web application interface with a dark gray background. At the top left, the text "Enter Your Name" is shown in a large, bold, white font. Below this, there are two input fields: "First Name" and "Last Name". The "First Name" field contains the text "Shreya" and the "Last Name" field contains "Karkera". To the right of these fields is a blue button with the text "SUBMIT" in white. In the bottom right corner, a dark gray modal dialog box is open. It has a title bar with a globe icon and the text "localhost:3002". The main content of the dialog says "Hello Shreya Karkera" in white text. At the bottom right of the dialog is a red button with the text "OK" in white.

EXERCISE 5: Build a simple React application that includes a Material UI button. When the button is clicked, it should toggle the visibility of a Material UI Card component

App.js

```
import React, { useState } from "react";
import Button from "@mui/material/Button";
import Card from "@mui/material/Card";
import CardContent from "@mui/material/CardContent";
import Typography from "@mui/material/Typography";

function App() {
  const [showCard, setShowCard] = useState(false);

  const toggleCard = () => {
    setShowCard((prev) => !prev);
  };

  return (
    <div style={{ display: "flex", flexDirection: "column", alignItems:
"center", marginTop: "50px" }}>
      <Button variant="contained" color="primary" onClick={toggleCard}>
        {showCard ? "Hide Card" : "Show Card"}
      </Button>
      {showCard && (
        <Card style={{ marginTop: "20px", width: "300px" }}>
          <CardContent>
            <Typography variant="h5" component="div">
              Material UI Card
            </Typography>
            <Typography variant="body2" color="text.secondary">
              This is a simple card that toggles visibility when you click
the button above.
            </Typography>
          </CardContent>
        </Card>
      )}
    </div>
  );
}
```

```
export default App;
```

OUTPUT:

SHOW CARD

HIDE CARD

Material UI Card

This is a simple card that toggles visibility when you click the button above.

PART B:

Exercise 1: Develop a React application that dynamically displays the capital city of a selected country. Implement the country selection dropdown as a separate, reusable component. The main application component should integrate this dropdown component and, upon selection of a country, display the corresponding capital city.

App.js

```
import React, { useState } from 'react';
import CountrySelector from './CountrySelector';

function App() {
  const [capital, setCapital] = useState('');

  const countryCapitalData = [
    { country: 'USA', capital: 'Washington, D.C.' },
    { country: 'India', capital: 'New Delhi' },
    { country: 'Germany', capital: 'Berlin' },
    { country: 'Japan', capital: 'Tokyo' },
    { country: 'China', capital: 'Beijing' },
  ];

  function handleCountrySelect(country) {
    const selectedCountry = countryCapitalData.find(item => item.country
=== country);
    if (selectedCountry) {
      setCapital(selectedCountry.capital);
    } else {
      setCapital('');
    }
  }

  return (
    <div>
      <h1>Country and Capital Viewer</h1>
      <CountrySelector onCountrySelect={handleCountrySelect} />
      {capital && <p>The capital city is: {capital}</p>}
    </div>
  );
}
```

```
);  
}  
  
export default App;
```

CountrySelector.js

```
import React from 'react';  
  
function CountrySelector({ onCountrySelect }) {  
  const countries = ['USA', 'India', 'Germany', 'Japan', 'China'];  
  
  return (  
    <div>  
      <label htmlFor="country">Select a country:</label>  
      <select id="country" onChange={(e) =>  
onCountrySelect(e.target.value)}>  
        <option value="">--Select--</option>  
        {countries.map((country) => (  
          <option key={country} value={country}>  
            {country}  
          </option>  
        ))}  
      </select>  
    </div>  
  );  
}  
  
export default CountrySelector;
```

OUTPUT:

Country and Capital Viewer

Select a country:

--Choose a country--

USA

India

Germany

Japan

Country and Capital Viewer

Select a country:

The capital city is: New Delhi

Exercise 2: Develop a react application and:

- i) Build your own Button component and render it three times. On user click, it should alert which button was clicked. (React JS).**

ii) Use the useState React hook to track how many times a button is clicked, and display the number